End of life challenges associatted with Polyoelefins (POs)

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Outline

1. Introduction
2. Reactivity of Alkene and Polyalkene
3. Challenges with End of Life
4. Conclusion
5. Appendix

Definition of Polyolefins (POs)

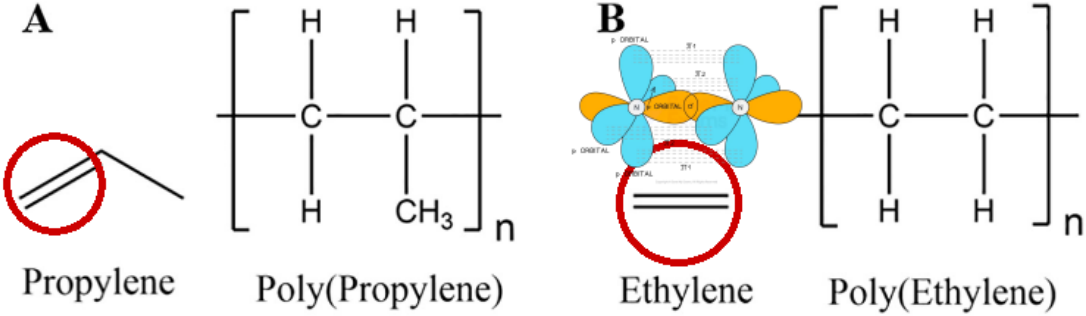


Figure 1. *Monomer and polymer of (A) PP and (B) PE [1].*

POs Represent 40% of the world plastic production

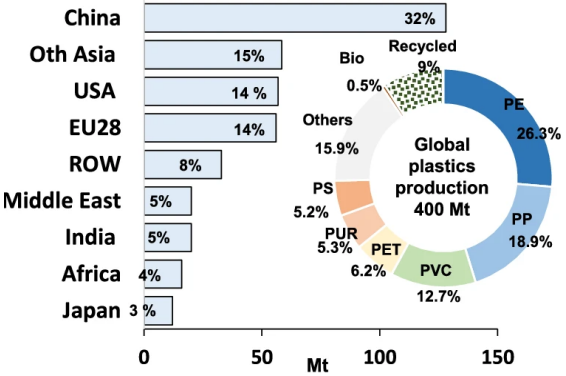


Figure 2. *Global plastic production breakdown by region/type [2]*

Why Are POs So Popular?

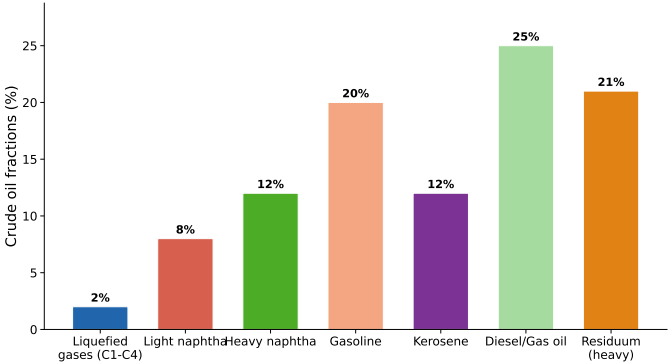


Figure 3. *Atmospheric distillation of crude oil [3]*

Naphtha → Ethylene, propylene **or** gasoline (oct. 95 or+) depending on the **Market**.

Because the Raw Materials Are Cheap

Naphtha + H_2O (steam) $\rightarrow$ Ethylene (30%) + Propylene (16%) + H_2 + CH_4 + C4s + BTX + coke

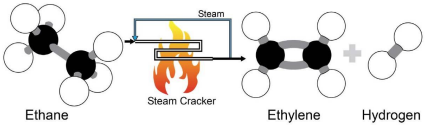


Figure 4. *Formation of ethylene from ethane (C_2H_6) [4]*

natural gas (95% CH_4 + 3% C_2H_6)

alternative source(shale gas of US Gulf Coast). [5]

Because POs Are Versatile

PP (Polypropylene)

- M_w
- Tacticity

PE (Polyethylene)

- M_w
- PDI
- Branching

for thermoplastics

M_w + PDI + branching + Tacticity $= \chi_c$ & properties (E,ρ, σ, μ and T_g)

trade off

PE density $\rightarrow \chi_c \rightarrow$ HDPE pipes vs LDPE films
PP tacticity $\rightarrow \chi_c \rightarrow$ PP ρ

POs exhibit excellent chemical resistance

Peu important	Alcanes et composés halogénés
	Éthers et thioéthers
↓	Alcènes et alcynes
	Amines
	Alcools et thiols
	Cétones
	Aldéhydes
	Nitriles
	Amides
	Esters
	Halogénures d'acide
	Acides carboxyliques
Très important	Cations

Figure 5. *Functional group reactivity [6].*

Summary of PP/PE properties

PE (Polyethylene) [7–9]

Properties

- Low density: 0.91–0.97 g/cm^3
- Chemical resistance: Excellent
- T_g = -120°C to -110°C
- T_m = 105 to 136°C
- χ_c = 0.4-0.9
- Temp range: -100°C to +130°C (HDPE), +80°C continuous
- Cost: $1,200–1,800/tonne (2026 avg)

PP (Polypropylene) [7–9]

Key Advantages

- Low density: 0.90–0.91 g/cm^3
- Chemical resistance: Excellent (acids, bases, oils)
- T_g = -120°C to -110°C
- T_m = 150 to 170°C
- χ_c = 0.3-0.6
- Temp range: 0°C to +140°C
- Cost: $1,100–1,600/tonne (2026 avg)

POs Are Mainly Used for Short-Term Applications

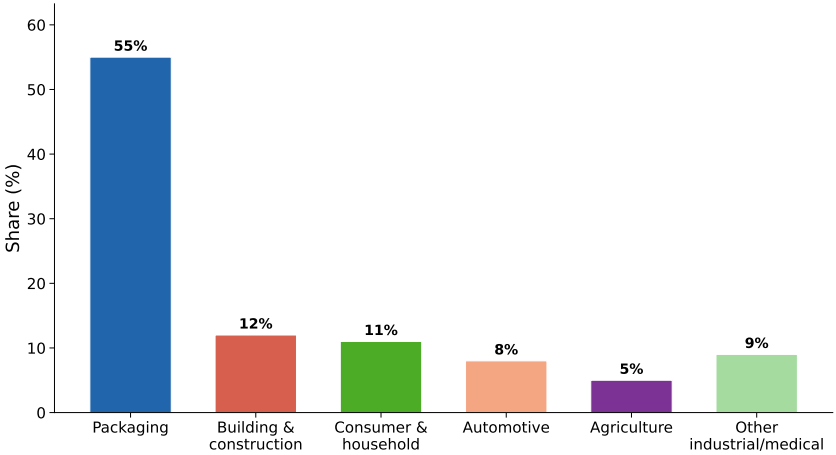


Figure 6. *POs per final use [10]*

Alkene's reactivity due to the π bond

typical reaction

- addition with a nucleophile
- **addition with a radical**
- Oxidation reaction

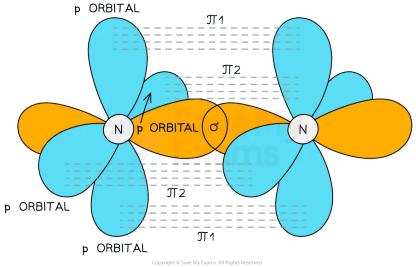


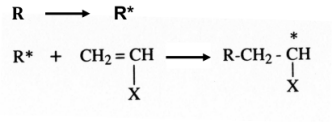
Figure 7. *π-bond [11].*

Alkenes can polymerize by Chain-reaction Polymerization [12]

- **Free Radical Polymerization (FRP)**
- **Complex coordination polymerization**

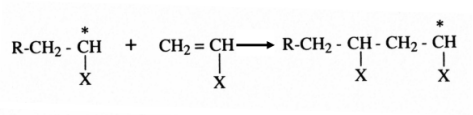
FRP : main reaction scheme [12]

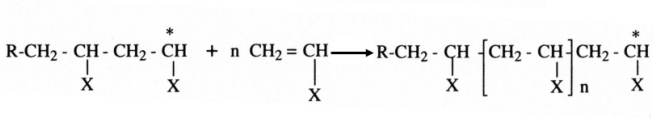
Initiation



R* = radical, anion or cation

Propagation





FRP: Parameters Influencing Stereochemistry

During FRP

Most vinyl polymers → **atacticity** due to the planar sp^2 radical center allowing random addition from either face [12].

Key parameters influencing stereoselectivity include: **T** & **steric hindrance**

PE & PP

PE is not concerned. **PP** is strictly atactic in FRP

Radical Transfer Between Two Chains Leads to Long Branches

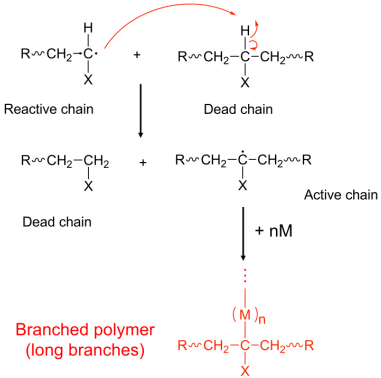


Figure 8. *Inter-chain transfer [13].*

→ long branches in PE.

Radical Transfer Within One Chain Leads to Short Branches

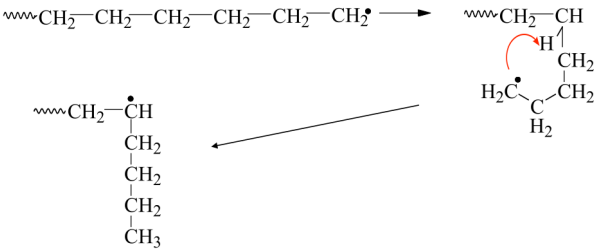


Figure 9. *Intra-chain transfer [13].*

→ short branches in PE.

Transfer to Monomer

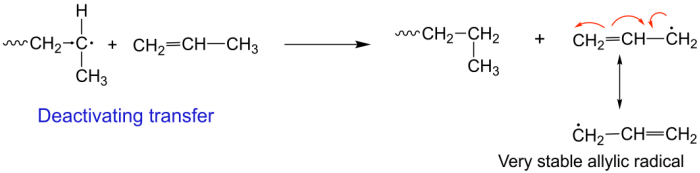


Figure 10. *Chain neutralization in PP [13].*

→ High molecular weight PP is not achievable in standard conditions.

FRP

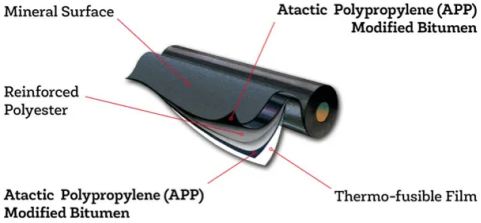
- atactic PP → $\chi = \pm 0 - 0.05$ [14].
- LDPE → $\chi = \pm 0.4 - 0.6$

FRP: Conclusion

Only LDPE

- High T and high P
- Amorphous ← a lot of branches of different sizes
- PDI = 7-20

Only atactic PP



Complex Coordination Polymerization: A Revolution

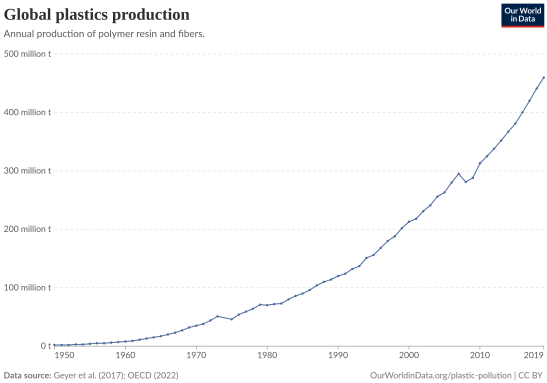


Figure 11. *Global plastic production since 1950 [15].*

Ziegler–Natta Catalyst

Catalyst:

- Heterogeneous transition metal (Ti)
- Organometallic of a metal from (I to III) (Al)

Typically:

$$TiCl_4 \text{ or } TiCl_3 + AlR_3 \tag{1}$$

with $R = CH - CH_3$ for polyethylene.

Major Results with the Introduction of the ZN Catalyst

According to Demoustier-Champagne *et al.* [12]

- Stereospecific polymerization of PP
- Synthesis of linear PE
- Copolymerization of PP/PE

→ Control of stereoregularity/stereochemistry and M_w through termination reactions.
→ Tacticity depends on the crystalline form of the catalyst.

Major Polyolefins Made by ZN Catalyst

PE

- HDPE ($\chi_c = 0.9$ & PDI = 3-5)
- UHWPE ($M_w = \pm 3 \times 10^6$)
- LLDPE (Ethylene copolymer with 5-12% of α-olefin)

Isotactic PP

- Soluble metallocene catalyst
- 99% tacticity

End-of-Life Routes for Polyolefins

- Mismanagement KO
- Mechanical recycling (MR) OK
- Biodegradability KO
- Chemical recycling (CR)OK (TRL low)
  - catalytic upcycling strategies (hydrogenolysis, (hydro)cracking, tandem processes involving metathesis, and selective oxidation) [16] OK
  - Pyrolysis (Thermal/Catalytic Cracking) [17]: Naphta OK
  - Gasification (production of syngas) OK
  - Depolymerization PE to PP OK
- Energy recovery OK
  - Incineration OK
  - Pyrolysis (Thermal/Catalytic Cracking) : fuel OK
  - Gasification (production of syngas) OK

MR: What Happens to the Polymer Structure?

According to Ragaert *et al.* [18] :

1. Re-extrusion step
2. The use phase

Degradation mechanisms:

- **Thermo-mechanical degradation.**
- Thermo-oxidative degradation.
- Hydrolytic degradation.
- **Impurities and contaminants.**

MR: Two Thermo-Mechanical Degradation Mechanisms

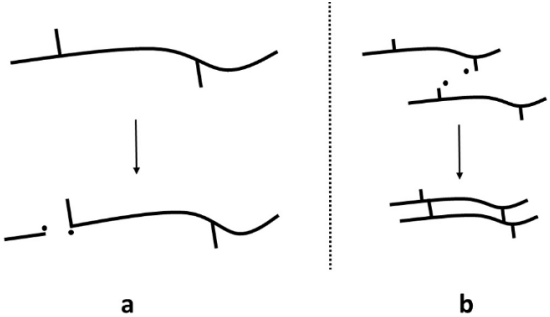


Figure 12. *Random chain scission (a) and cross-linking (b) [18].*

$M_w \downarrow$ **or** $M_w \uparrow$ and $M_w \propto E$, σ_Y, ϵ_{break}, σ_{Impact}, χ_c and μ.

MR: Polyolefins Behave Very Differently During Mechanical Recycling

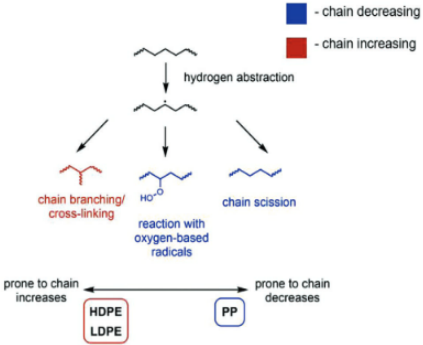


Figure 13. *General reaction scheme of polyolefins undergoing extrusion [18].*

MR: Effect of Recycling on Polyolefin Properties

Property	Typical Effect	HDPE	PP
Molecular Weight	↓ (chain scission)	vary	↓ Significant
PDI	Changes	↑	↓
Tensile Strength	↓ Loss	10–30%	10–30%
Crystallinity	↓ Loss	5–10%	5–10%

Table 1. *

Loss of properties [1]

MR/CR: Sorting Problem and PE/PP Blend Incompatibility

PE/PP overlap

- PP/PE have $\pm$ same ρ
- PP/PE have same optical properties
- immiscible

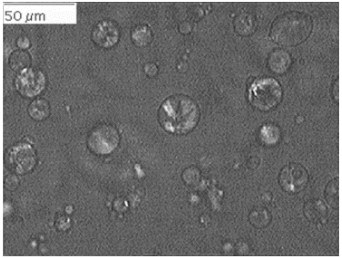


Figure 14. *TEM image : PP crystal into LDPE [19].*

CR: main hurdles

TRL Levels

- Pyrolysis (TRL +++) [17]
- Gasification (TRL ++) [17]
- Depolymerization (TRL+) [16]

Main hurdles

- High energy consumption
  - Brent Crude ≈ 110.4 \$/barrel
  - Carbon Tax
- Sorting
  - catalyst poisoning

Others: Challenges and Perspectives

Key processing constraints for rPE

- Process tolerance varies — thin films are less forgiving.
- Common contaminants: paper fibre, other polymers, rubber.
- Branched gels in rPE limit stretchability.
- Optical quality: many rPE grades are greyish vs. virgin PE.

Conclusion

POs are popular due to

- Cheap raw material
- Processable
- Versatile
- Stable

Good processability due to ZN catalyst

- control of M_w & PDI
- control of stereochemistry

EoL

low price of RwM + purity of polymer blend + chain degradation & high E requirement

Thank you for your attention

Do you have any questions ?

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Bibliography I

1. Jubinville, D. *et al.* A comprehensive review of global production and recycling methods of polyolefin (PO) based products and their post-recycling applications. *Sustainable Materials and Technologies* **25,** e00188 (Sept. 1, 2020).
2. Houssini, K. *et al.* Complexities of the global plastics supply chain revealed in a trade-linked material flow analysis. *Commun Earth Environ* **6,** 257 (Apr. 10, 2025).
3. *Crude oil | Definition, Characteristics, & Facts | Britannica.* `https://www.britannica.com/science/crude-oil` (2026).
4. *Clendenin, West Virginia: Landmark for the birthplace of the petrochemical industry* American Chemical Society. `https://www.acs.org/education/whatischemistry/landmarks/petrochemical-industry-birthplace.html` (2026).

Bibliography II

5. *Composition of natural gas: understanding its key elements.* https://met.com/en/media/energy-insight/composition-of-natural-gas/ (2026).
6. Demoustier-Champagne, S. & Fustin, C.-A. *LMAPR1230 Chimie Organique.* Course. Course. Louvain-la-neuve, 2023.
7. Hughes, C. *Polyethylene vs Polypropylene: What's the Difference?* Renegade Plastics. https://renegadeplastics.com/blog/polyethylene-vs-polypropylene-whats-the-difference/ (2026).
8. SpecialChem. *Glass Transition Temperature (Tg) of Plastics.* https://www.specialchem.com/plastics/guide/glass-transition-temperature (2026).
9. *Glass transition.* https://en.wikipedia.org/wiki/Glass_transition (2026).

Bibliography III

10. Polyolefins Circular Economy Platform (PCEP). *Plastic Pledge Statement: PCEP – a Key Platform to Drive Polyolefins Circularity*. Accédé le 14 mars 2026. Sept. 2018.
11. *Sigma and Pi Bonds - IB Chemistry Revision Notes* Save My Exams. https://www.savemyexams.com/dp/chemistry/ib/23/hl/revision-notes/models-of-bonding-and-structure/the-covalent-model/sigma-and-pi-bonds/ (2026).
12. Demoustier-Champagne, S. *et al.* *LMAPR2019 Polymer Sciences and engineering*. Course. Course. Louvain-la-neuve, 2024.
13. De Moustier, S. *LMAPR2019 Polymer Sciences and engineering*. Course. Course. Louvain-la-neuve, 2024.
14. *Polypropylene*. https://en.wikipedia.org/wiki/Polypropylene (2026).
15. Ritchie, H. *et al.* Plastic Pollution. *Our World in Data*. https://ourworldindata.org/plastic-pollution (2023).

Bibliography IV

16. Garland, C. P. *et al.* Catalytic Upcycling of Polyolefins. *Chemical Reviews* **124,** 10234–10289 (2024).
17. Frączak, D. in *Waste Material Recycling in the Circular Economy - Challenges and Developments* (IntechOpen, Aug. 5, 2021).
18. Ragaert, K. *et al.* Mechanical and chemical recycling of solid plastic waste. *Waste Management* **69,** 24–58 (Nov. 2017).
19. Shanks, R. A. *et al.* Polypropylene–polyethylene blend morphology controlled by time–temperature–miscibility. *Polymer* **41,** 2133–2139 (Mar. 1, 2000).

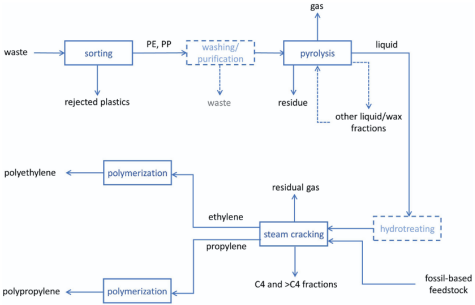

Figure 3.
Scheme of polyethylene and polypropylene from the plastic waste production process.

Figure 15. *Flow sheet of pyrolysis [17].*

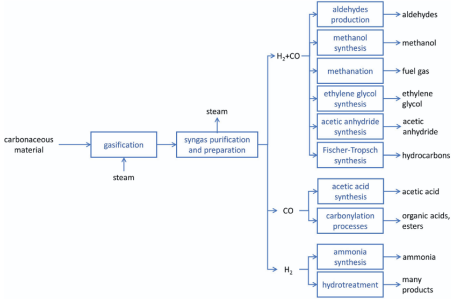

Figure 5.
Overview of potential syngas applications [28].

Figure 16. *Flow sheet of syngaz production [17].*

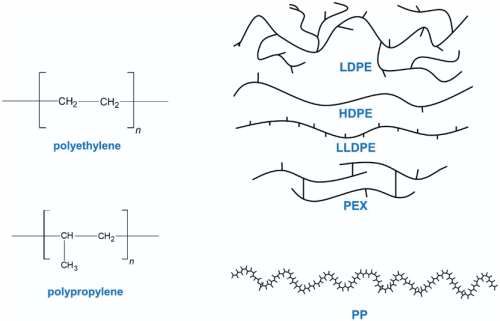

Figure 17. *Structure of PP/PE [17].*